

Q1 - 26 July - Shift 1

In young's double slit experiment, the fringe width is 12mm. If the entire arrangement is placed in water of refractive index $\frac{4}{3}$, then the fringe width becomes (in mm)

- (A) 16 (B) 9
(C) 48 (D) 12

Space for your notes:

Q2 - 27 July - Shift 1

Two beams of light having intensities I and $4I$ interfere to produce a fringe pattern on a screen. The phase difference between the two beams are $\pi/2$ and $\pi/3$ at points A and B respectively. The difference between the resultant intensities at the two points is xI . The value of x will be _____.

Space for your notes:

Q3 - 27 July - Shift 2

Two coherent sources of light interfere. The intensity ratio of two sources is 1 : 4. For this interference pattern if the value of $\frac{I_{\max} + I_{\min}}{I_{\max} - I_{\min}}$ is

Space for your notes:

equal to $\frac{2\alpha + 1}{\beta + 3}$, then $\frac{\alpha}{\beta}$ will be :

- (A) 1.5 (B) 2 (C) 0.5 (D) 1

Q4 - 28 July - Shift 1

In a Young's double slit experiment, a laser light of 560 nm produces an interference pattern with consecutive bright fringes' separation of 7.2 mm. Now another light is used to produce an interference pattern with consecutive bright fringes' separation of 8.1 mm. The wavelength of second light is _____ nm.

Space for your notes:

Q5 - 29 July - Shift 2

An unpolarised light beam of intensity $2I_0$ is passed through a polaroid P and then through another polaroid Q which is oriented in such a way that its passing axis makes an angle of 30° relative to that of P. The intensity of the emergent light is

- (A) $\frac{I_0}{4}$ (B) $\frac{I_0}{2}$ (C) $\frac{3I_0}{4}$ (D) $\frac{3I_0}{2}$

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Answer Key**Q1 (B)****Q2 (2)****Q3 (B)****Q4 (630)****Q5 (C)**

#MathBoleTohMathonGo

Q1 (B)

For a given light wavelength corresponding a medium of refractive index μ

$$\lambda_{med} = \frac{\lambda_{vacuum}}{\mu}$$

and we know that fringe width $\beta = \frac{\lambda D}{d}$

$$\text{Therefore, } \beta_{med} = \frac{\beta_{vacuum}}{\mu} = \frac{12}{\frac{4}{3}} = 9 \text{ mm}$$

Q2 (2)

$$\phi_A = \frac{\pi}{2}$$

$$\phi_B = \frac{\pi}{3}$$

$$I_A = I + 4I + 2\sqrt{I}\sqrt{4I} \cos\left(\frac{\pi}{2}\right)$$

$$= 5I + 4I(0) = 5I$$

$$+ 4I + 2\sqrt{I}\sqrt{4I} \cos(60^\circ)$$

$$- 4I \times \frac{1}{2} = 7I$$

$$I = 7I - 5I = 2I, (x = 2)$$

Q3 (B)

$$\frac{I_1}{I_2} = \frac{1}{4}$$

$$I_2 = 4I_1$$

$$I_{\max} = I_1 + 4I_1 + 2\sqrt{I_1 \cdot 4I_1} = 9I_1$$

$$I_{\min} = I_1 + 4I_1 - 2\sqrt{I_1 \cdot 4I_1} = I_1$$

$$\therefore \frac{9I_1 + I_1}{9I_1 - I_1} = \frac{10}{8} = \frac{5}{4} = \frac{2\alpha + 1}{\beta + 1}$$

$$\alpha = 2 \quad \beta = 1$$

$$\therefore \frac{\alpha}{\beta} = \frac{2}{1} = 2$$

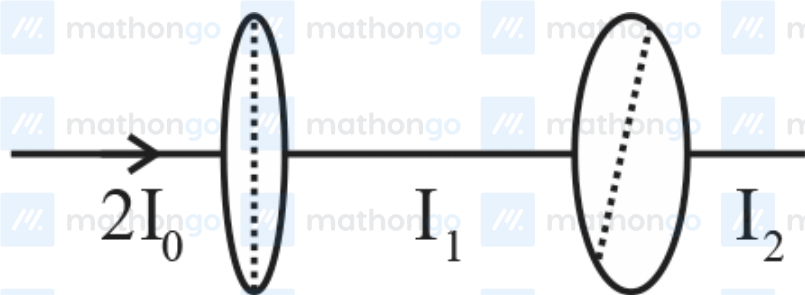
Q4 (630)

$$\beta \propto \lambda$$

$$\lambda_2 = \frac{9}{8} \lambda_1$$

$$\therefore \beta_2 = \frac{9}{8} \beta_1 = \frac{9}{8} \times 560 = \boxed{630} \text{ nm}.$$

Q5 (C)



$$I_1 = \frac{1}{2}(2I_0) = I_0$$

$$I_2 = I_1 \cos^2 30^\circ$$

$$= I_0 \cdot \frac{3}{4} = \frac{3I_0}{4}$$